Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/29/2026
PI Name:	Lu, Xiao-Hong Ph.D.		
Project Title:	Develop a Novel Single-Cell Biodosimetry for Brain Genomic Instability and Neurodegeneration to Predict Clinical Health Outcomes in Human Spaceflight Crews		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Shreveport	**State:**	LA
Zip Code:	71103	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	11/15/2020	**End Date:**	12/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per F. Hernandez/ARC (Ed., 7/2/25) NOTE: End date changed to 06/01/2025 per F. Hernandez/ARC (Ed., 2/12/25) NOTE: End date changed to 11/30/2024 per NSSC information (Ed., 11/15/23) NOTE: End date is 11/30/2023 (incorrectly listed in NSSC as 11/14/2021) per F. Hernandez/ARC (Ed., 7/27/21)		
Key Personnel Changes/Previous PI:	Per the PI, the list of Co-Investigators has changed. During the past reporting period, the team has been working with the newly established bioinformatics center at Louisiana State University for the project, and Dr. Cvek was not involved. Dr. Harrison has moved to a new position and will no longer serve on this project. (Ed., 9/17/23)		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0273		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As NASA plans future exploration missions to the Lunar and Martian surfaces, realistic ground-based analog studies and more predictive biodosimetry are needed to assess whether the space radiation poses a detrimental risk of brain genomic instability and neurodegeneration that leads to late-onset behavioral deterioration for spaceflight crews. Implementing a recently developed method of recreating the intravehicular (IVA) radiation environment expected on spaceflight vehicles and habitats and a novel genetic sensor, this proposal addresses Research Topic 3 – Animal Biology Studies in support of Human Space Exploration and Sub-topic AB1-A – Behavior and underlying neural function in Appendix D: Solicitation of Proposals for Flight and Ground Space Biology Research. We propose to determine how the space environment and sex affect brain genomic stability and consequent age-related brain structure and function changes. Our studies will support Human Space Exploration, by contributing the first biodosimetry for quantifying brain DNA instability and neurodegenerative changes to predict clinical health outcomes in human spaceflight crews and the utility of available ground-based analogs to realize basic mechanisms that can lead to the development of biologic counter-measures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies will support Human Space Exploration, by contributing the first biodosimetry for quantifying brain DNA instability and neurodegenerative changes to predict clinical health outcomes in human spaceflight crews and the utility of available ground-based analogs to realize basic mechanisms that can lead to the development of biologic counter-measures.
Task Progress:	The long-lasting effects of Galactic Cosmic Radiation (GCR) exposure on brain health must be examined as humanity prepares for extended missions into deep space. To model such conditions, our lab has utilized mice exposed to ground-based GCR analogues at lunar- and Mars-mission-relevant doses, administered as either 5-ion simplified GCR simulation (sGCR), or 1000 MeV/n 56Fe passed through a moderator block to model intravehicular spallation. We here utilize a novel genetic sensor, PRISM, to map the acute and chronic impact of exposure to deep space radiation with sensitive spatiotemporal in vivo visualization of GCR-induced genomic instability. This genetic reporter revealed a largely dose-independent increase in activation of the DNA Damage Response (DDR) in neurons of the cortex, striatum, and hippocampus of mice irradiated with simulated GCR, indicating that neurons are highly susceptible to DNA damage at mission-relevant exposure levels. Furthermore, reactive microgliosis (increased cell size and counts of Iba1-expressing microglia) was observed in multiple brain regions – including the prefrontal cortex, striatum, and hippocampus at 2, 4.5, and 18 months post-exposure, with an age-dependent increase in neuroinflammation. Finally, our extensive behavioral assessments ranging from 2- to 18-months post-irradiation revealed persistent deficits – including reduced sensorimotor capacity, changes in novelty-seeking and risk-taking behaviors, early emotional deficits, and working memory deterioration with age. We also observed notable gender differences in these behavioral deficits among both young and aged mice. Our findings indicate that simulated GCR exposure has a variety of negative impacts on brain health, including persistent activation of neuronal DNA damage response and chronic neuroinflammation, which manifest as long-lasting cognitive, emotional, and sensorimotor deficiencies.
Bibliography Type:	Description: (Last Updated: 09/26/2024)